

SECTION 6 – INTERCONNECT DATA

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6-1 INTRODUCTION

Section 6, INTERCONNECT DATA, addresses system interconnections which are impacted by the various Horizon LX upgrade catalogs. It's subsections are broken down as follows:

- 6-1 INTRODUCTION
 - general upgrade information, component designations
- 6-2 SYSTEM INTERCONNECTS ADDED BY UPGRADES
 - cable interconnects for the system provided by Signa LX With EXCITE Technology Upgrade catalogs
 - cable interconnects for the system provided by Signa LX with ACGD Upgrade catalogs
- 6-3 SIGNA CVMR ADVANTAGE WORKSTATION INTERCONNECTS
 - cable interconnects for Signa CVMR Option Advantage Workstation Package
- 6-4 INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES
 - existing system cable interconnects re-labeled and re-routed during Horizon LX with ACGD Upgrades
- 6-6 SYSTEM INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES
 - existing system cable interconnects removed during Horizon LX Upgrades
- 6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE
 - cables interconnects removed and discarded during upgrade to Signa Horizon LX Computer only configuration with no gradient subsystem change
- 6-8 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
 - need to plan Rear Pedestal cable access and routing due to increase of cabling and hoses
- 6-9 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for systems with ACGD Cabinet
- 6-10 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD
 - additional cable interconnects for the system provided by Spectroscopy Option Upgrade catalogs for systems with SGD Cabinet

Note

Magnet subsystem interconnects are not impacted by Horizon upgrades and therefore are not listed in this section.

6-1-1 Component Designators

The Component Designator System is used as the means of identifying system components in a consistent manner. All subsystem cabinets and other components are referred to by their component designators in the diagrams and tables of this section. (For example, the System Cabinet is referred to as MR2.) Refer to Table 6-1 for component designators impacted by Signa Horizon LX upgrades.

TABLE 6-1
COMPONENT DESIGNATION

APPLICABLE SYSTEM	COMPONENT DESIGNATOR	DESCRIPTION
Signa LX with ACGD	MG2	Magnet Enclosure
	MG3	Magnet Rear Pedestal
	MG6	Blower Box
	MR1	SRF Cabinet
	MR2	System Cabinet
	MR2	System Cabinet for Signa LX With EXCITE Technology or Signa LX
	MR3	ACGD/PDU Cabinet has PD1 in lower portion cabinet
	MS4	Magnet Rundown Unit
	PD1	Power Distribution Unit (PDU) is a module in lower portion of ACGD/PDU Cabinet
	PP1	Penetration Panel
Signa LX with ACGD	WC1	Gradient Coil Water Chiller
1.5T System Option	MNS	MNS Cabinet
System CVMR Option	AW	Advantage Workstation
Signa Horizon LX with SGD, Signa Horizon, Signa Advantage or Signa Obsolete Equipment	CC1	Computer Cabinet
	CC2	Auxiliary Computer Cabinet
	CM1	Cardiac Monitor
	HE1	Heat Exchanger
	MR1	RF/PDU Cabinet RF/Penetration Cabinet RF Cabinet
	MR3	SGD Base Gradient Cabinet for Base System or SGD Hi-Slew Gradient Cabinet for HiSpeed or EchoSpeed 8645 Gradient Cabinet Gradient Amp Cabinet
	MR4	Gradient Resistor Box (may exist on Pre-Performance Plus systems)
	MR7	GRAM Cabinet
	OC1	Operator Console
	PD1	Power Distribution Unit (PDU) is a module in lower portion of RF/PDU Cabinet Full Size (150 KVA or 75 KVA) or Compact Power Distribution Unit
	PP2	Penetration Cabinet

6-1-2 Definition of Terms

The definition of terms used in following sections tables are:

- Group Number: identifying number referenced to bundles (i.e. groups) of cables as shown in interconnect diagrams
- Between Units (From/To): component designators as found in tables
- Usable Length: total length of a cable MINUS any required take up within cabinets
- Run Number: unique number assigned to each GE-supplied cable

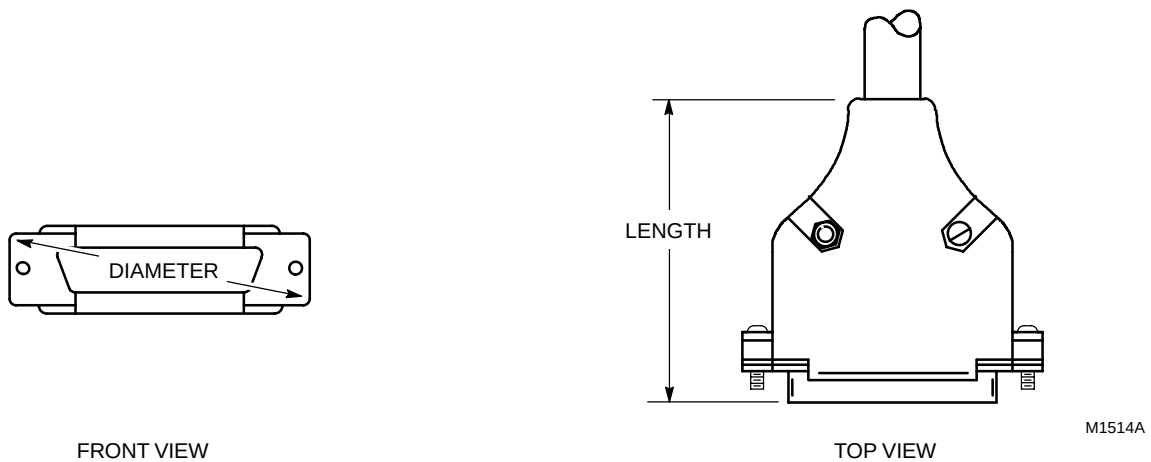
Note

The run number must be used when making special cable orders.

- Cable Diameter: diameter of an individual cable
- Plug Pulling Diameter x Length: cable plug dimensions

Note

In some cases, a cable has more than one connector on an end. These cables will have the number of connectors following the english dimensions of the plug pulling diameter times length (e.g. '2.0x3.25 -x2' means there are 2 connectors with dimensions of 2.0 in. diameter and 3.25 in. length). Of course, the same number of connectors apply to the metric dimensions as well.



SUBMINIATURE-D CONNECTOR PLUG PULLING DIMENSIONS
ILLUSTRATION 6-1

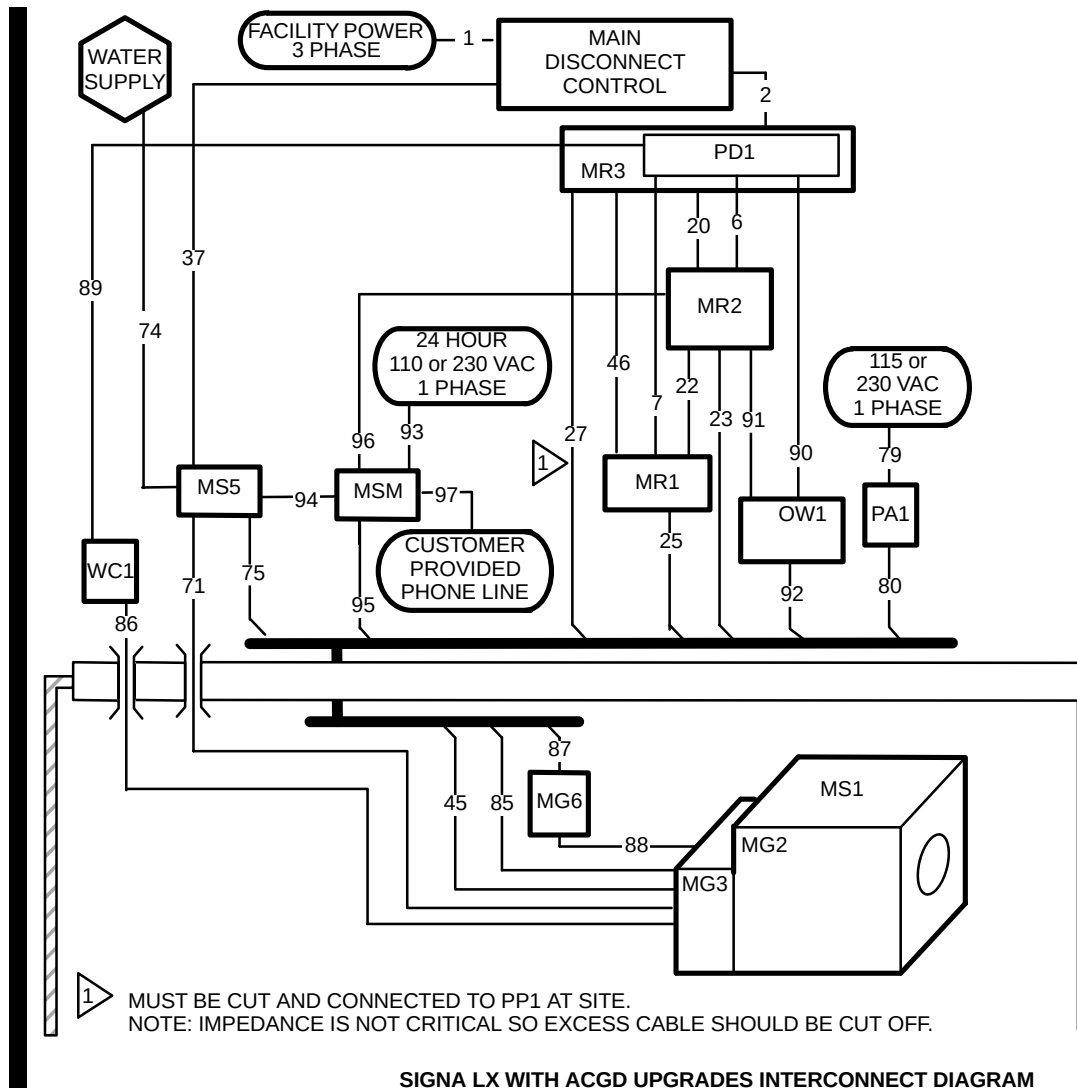
6-2 SYSTEM INTERCONNECTS ADDED BY ACGD UPGRADES

6-2-1 Introduction

Illustration 6-2 interconnect diagrams shows the groups to which cables are added by the various Signa LX upgrade catalogs. The following sub-sections list the cables provided by the various Signa LX upgrade catalogs. Conduit or pipe is not recommended for cable runs since the system uses many prefabricated cables with large connectors. Unless otherwise specified, cables and components listed in tables in this section are supplied by GE.

Note

Need to verify customer supplied Group1 (Facility Power to Main Disconnect Control) and Group 2 (Main Disconnect Control) are properly sized for specifications in Section 5, Power Requirements.



6-2-2 M1033L – Indigo to Octane II with EXCITE Upgrade

M1033L provides interconnects for an Indigo to Octane II with EXCITE Upgrade listed in Table 6-2.

TABLE 6-2
CABLES PROVIDED BY M1033L

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
91	MR2	OW1	51 (15.5)	1032	0.525 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-3 M1060JW – Shield/Cryo Cooler Compressor

M1060JW provides interconnects for Shield/Cryo Cooler Compressor for the LCC Magnet listed in Table 6-3.

TABLE 6-3
CABLES PROVIDED BY M1060JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
37	MDP	MS5	29.8 (9.1) minus takeup at MDP	—	1.29 (32.75)	Hard Wired	Hard Wired	Both ends have liquid tight connector which wires pass through for hard wire connections.
45	PP1	MG3	30 (9.15)	624	0.26 (6.6)	Ring Terminals	1.00x1.38 (25.4x35.1)	
71	MS5	MS1	38.5 (11.7)	621 622	1.65 (41.9) 1.65 (41.9)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3) 2.00x3.75 (50.8x95.3)	Runs 621 and 622 are continuous helium lines routed through PP1. Bend radius of Runs 621 and 622 is 8 in. (203.2 mm). Cable diameter includes foam insulation installed on lines.
75	MS5	PP1	42 (12.8)	623	0.21 (5.28)	1.00x1.38 (25.4x35.1)	Ring Terminals	

6-2-4 M1060LC – 1.5T CXK4 Magnet Upgrade with Wide Open Enclosure

M1060LC provides interconnects for the LCC Magnet listed in Table 6-4.

TABLE 6-4
CABLES PROVIDED BY CATALOGS M1060LC & M1860LC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	828	0.31 (7.75)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				829	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
48	MS4	MS1	86 (26.2)	606	0.30 (7.6)	0.65x1.85 (16.5x47.0)	0.65x1.85 (16.5x47.0)	
93	24 Hour Facility Power	MSM	6 (1.8) minus takeups at each end	—	0.375 (9.5)	1.2x3.00 (30.5x76.2)	1.25x2.5 (31.8x63.5)	
94	MSM	MS5	54 (16.4) minus takeup at MSM	826	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
95	MSM	PP1	54 (16.4) minus takeup at MSM	824	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				825	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
96	MSM	MR2	54 (16.4) minus takeup at MSM	823	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

6-2-5 M1090JP & M1090JR – Enclosure Upgrade For Fixed Site

The enclosure catalogs for fixed sites provide the Blower Box to Magnet Enclosure air hoses. Refer to Table 6-5 for details.

TABLE 6-5
CABLES PROVIDED BY CATALOGS M1090JP, M1090JR, & M1191JF

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
88	MG6	MG3	50 (15.2) minus takeup, See Notes	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The 6.5 in. (165.1 mm) hose has bend radius of 4 in. (101.6 mm). The 4.5 in. (104.3 mm) hose has bend radius of 2 in. (50.8 mm). The flexible vinyl hoses are cut to length during installation. Total length provided is 50 ft (15.2 m). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				—	4.5 (104.3)	Flexible vinyl hose	Flexible vinyl hose	

6-2-6 M1090JW – Laser to Halogen Alignment Lights Upgrade

M1090JW provides interconnect for the Halogen Alignment Lights listed in Table 6-4.

TABLE 6-6
CABLES PROVIDED BY CATALOG M1090JW

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	385	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	

6-2-7 M1090PC – 5.5 Phased Array Compatibility Kits for 1.5T System

M1090PC provides interconnects for Signa Horizon LX 1.5T System without Phased Array. Refer to Table 6-7 interconnect details.

TABLE 6-7
CABLES PROVIDED BY M1090PC, M1090PH, M1890PA, M1890PH

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
23	MR2	PP1	31 (19.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
45	PP1	MG3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	

6-2-8 M1090PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.5T System

M1090PH provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalog M1090PH are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-9 M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) For LX Systems Without Phase Array

M1090PW provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalog M1090PW are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-10 M1090SJ – 5.5 To LX Cable Upgrade For Fixed Site

Catalog M1090SJ provides fixed site system interconnect cables for 5.5 Signa Horizon configuration to Signa Horizon LX. Also included in this catalog are Horizon LX upgrade power and ground cables. Refer to Table 6-14 for M1090SJ cables details.

TABLE 6-8
CABLES PROVIDED BY CATALOG M1090SJ

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals	Ring Terminals	Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				790	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-11 M1191JF – Enclosure Upgrade For Transportable/Relocatable

Cables provided by catalog M1191JF are the same as catalogs M1090JP and M1090JR. Refer to Table 6-5 for M1191JF cables details.

6-2-12 M1191JM – 5.5 To LX Cable Upgrade for True Mobile with Rear Mounted Penetration Panel

All signal cables are replaced during a True Horizon LX Mobile upgrade, PDU to cabinet power cables are retained and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-13 M1191JN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade, PDU to cabinet power cables are retained and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-14 M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet

M1191SN provides interconnects for the 1.5T LCC Magnet listed in Table 6-9.

TABLE 6-9
CABLES PROVIDED BY CATALOGS M1191SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	235	0.82 (20.8)	1.44x2.00 (36.6x50.8)	1.44x2.00 (36.6x50.8)	Used for 1.5T system only. Has a minimum bend radius of 10 in. (254 mm).
				236	0.59 (15.0)	0.92x2.10 (23.4x53.3)	2.00x2.10 (50.8x53.3)	Used for 1.5T system only. Has a minimum bend radius of 5 in. (127 mm).
27	MR3	PP1	33 (10.0)	762	1.21 (30.7)	Hard wired	Ring Terminals	These runs are cut and connected to PP1 at site; total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85 for Magnet Room routing of these interconnects.
				763	1.21 (30.7)	Hard Wired	Ring Terminals	
				764	1.21 (30.7)	Hard Wired	Ring Terminals	
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
45	PP1	MG2/3	55 (16.76)	257	1.10 (27.9)	1.40x3.40 (35.6x86.4)	1.40x3.40 (35.6x86.4)	Used for 1.5T system only. Has a minimum bend radius of 10 in. (254 mm).
			55 (16.76)	258	0.64 (16.3)	0.90x2.50 (22.9x63.5)	0.90x2.50 (22.9x63.5)	Used for 1.5T system only. Has a minimum bend radius of 5 in. (127 mm).
				262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-2-14 M1191SN – Collector Kit for Upgrade to 1.5T LCC Magnet

TABLE 6-9 (Continued)
CABLES PROVIDED BY CATALOGS M1191SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	55 (16.76)	700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. For Equipment Room routing of these interconnects refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	065	0.635 (16.1)	Hard Wired	Hard Wired	#10 AWG / 3 wire Power cable

6-2-15 M1191SP – Collector Kit for Upgrade to 1.5T LCC Magnet

M1191SP provides interconnects for the LCC Magnet listed in Table 6-10.

TABLE 6-10
CABLES PROVIDED BY CATALOGS M1191SP & M1891SP

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
88	MG6	MG3	13 (3.95) minus takeup	—	6.5 (165.1)	Flexible vinyl hose	Flexible vinyl hose	The flexible vinyl hoses are cut to length during installation.

6-2-16 M1860LC – 1.0T CXK4 Magnet Upgrade with Wide Open Enclosure

Cables provided by catalog M1860LC for 1.0T system are the same as catalog M1090PC for 1.5T system. Refer to Table 6-4 for cables details.

6-2-17 M1890PA – 5.5 Phased Array Compatibility Kits For 1.0T System

Cables provided by catalog M1890PA for 1.0T system are the same as catalog M1090PC for 1.5T system. Refer to Table 6-7 for M1890PA cables details.

6-2-18 M1890PH – Pre LX to LX Without Phase Array to 4 channel Phase Array Adder Kit For 1.0T System

M1890PH provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalogs M1090PH are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-19 M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array

M1890PW provides interconnects for Signa Horizon LX 1.5T System without Phased Array upgrade to 4 channel Phased Array. Cables provided by catalogs M1090PW are the same as catalog M1090PC. Refer to Table 6-7 for M1090PH cables details.

6-2-20 M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet

M1191SN provides interconnects for the 1.0T LCC Magnet listed in Table 6-11.

TABLE 6-11
CABLES PROVIDED BY CATALOGS M1891SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	745	0.50 (12.7)	0.91x2.50 (23.0x63.5)	0.91x2.50 (23.0x63.5)	Used for 1.0T system only.
				748	0.44 (11.0)	0.63x2.01 (15.9x51.1)	0.63x2.01 (15.9x51.1)	Used for 1.0T system only.
27	MR3	PP1	33 (10.0)	762	1.21 (30.7)	Hard wired	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85 for Magnet Room routing of these interconnects.
				763	1.21 (30.7)	Hard Wired	Ring Terminals	
				764	1.21 (30.7)	Hard Wired	Ring Terminals	
42	PP1	E01	84 (25.6) minus takeup at EO1	297	0.35 (8.9)	1.30x2.00 (33.5x50.8)	Hard Wired	
45	PP1	MG2/3	55 (16.76)	262	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				282	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				318	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				387	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				(Continued)				

6-2-14 M1891SN – Collector Kit for Upgrade to 1.0T LCC Magnet

TABLE 6-11 (Continued)
CABLES PROVIDED BY CATALOGS M1891SN

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45 (contin- ue)	PP1	MG2/3	55 (16.76)	727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for 1.0T system only. Used for 1.0T system only.
				746	0.50 (12.7)	0.91x2.50 (23x63.5)	0.91x2.45 (23x62.2)	
				747	0.440 (11.2)	0.63x2.01 (15.9x51.1)	0.63x2.01 (15.9x51.1)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
78	RF COM- MON GND STUD	MS1 GND STUD	60 (18.29) minus takeup at RF Common GND Stud	040	0.464 (11.79)	Hard Wired	Ring Terminal	1 ground wire.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. For Equipment Room routing of these interconnects refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	065	0.635 (16.1)	Hard Wired	Hard Wired	#10 AWG / 3 wire Power cable

6-2-21 M1891SP – Collector Kit for Upgrade to 1.5T LCC Magnet

Cables provided by catalog M1891SP for 1.0T system are the same as catalog M1091SP for 1.5T LCC Magnet. Refer to Table 6-10 for cables details.

6-2-22 M3090BA – 1.0T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Catalog M3090BA provides ACGD upgrade power and ground cables. Refer to Table 6-13 for cables details.

TABLE 6-12
CABLES PROVIDED BY CATALOG M3090BA, M3090BB, & M3090BC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-2-23 M3090BB – ACGD Upgrade for system with SGD Gradient Cabinet & RF/PDU Cabinet

Cables provided by catalogs M3090BB are the same as catalogs M3090BA. Refer to Table 6-13 for cables details.

6-2-24 M3090BC – 1.5T ACGD Upgrade for system with SGD Gradient Cabinet & RF/Penetration Cabinet

Cables provided by catalogs M3090BB are the same as catalogs M3090BA. Refer to Table 6-13 for cables details.

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CA provides fixed site system interconnect cables for Signa Advantage 4.X configuration to Signa Horizon LX with ACGD. Also included in this catalog are Horizon LX with ACGD upgrade power and ground cables, fiber optic cables, and the Gradient Coil Water Cooling flexible water lines. Refer to Table 6-13 for cables details.

TABLE 6-13
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wire
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wire
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	41 (12.50)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				773	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6–8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				715	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				716	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
			pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	Pneumatic Patient Alert connection, refer to Group 80 for route.	
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continu- ously routed from PA1 through PP1 and MG3 to MG3.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6–8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
(Continued)								

6-2-25 M3090CA – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-13 (Continued)
CABLES PROVIDED BY CATALOG M3090CA

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CB provides fixed site system interconnect cables for Signa Advantage 5.1-5.4 configurations to Signa LX with ACGD. Also included in this catalog are Horizon LX with ACGD upgrade power and ground cables, fiber optic cables, and the Gradient Coil Water Chiller flexible water lines. Refer to Table 6-14 for cables details.

TABLE 6-14
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
(Continued)								

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-14 (Continued)
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
27	MR3	PP1	31 (9.45)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 85. These cables have a minimum bend radius of 8 in. (203 mm).
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				778	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				779	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				780	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				781	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				782	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	Pneumatic Patient Alert connection, refer to Group 80 for route.
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
85	PP1	MG3	30 (9.15)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	These runs are cut and connected to PP1 at site: total length is 100 ft (30.5 m) including both Equipment and Magnet Rooms. Refer to Group 27. These cables have a minimum bend radius of 8 in. (203 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	

(Continued)

6-2-26 M3090CB – 5.1-5.4 To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-14 (Continued)
CABLES PROVIDED BY CATALOGS M3090CB & M3090CC

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
86	WC1	MG2	80 (24.4)	536	0.75 (19.1)	Flexible Tubing	Flexible Tubing	Runs 536 and 537 are routed through waveguides in PP1.
				537	0.75 (19.1)	Flexible Tubing	Flexible Tubing	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
				784	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
				786	0.64 (16.3)	Ring Terminals	2.28x3.85 (57.9x97.8)	
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-27 M3090CC – 5.5 Software Only Site To LX With ACGD Cable Upgrade For Fixed Site

Cables provided by catalog M3090CC is the same as catalog M3090CB. Refer to Table 6-14 for cables details.

6-2-28 M3090CD – 5.5 To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CD provides fixed site system interconnect cables for 5.5 Signa Horizon configuration to Signa Horizon LX with ACGD Gradient. Also included in this catalog are Horizon LX upgrade power and ground cables. Refer to Table 6-15 for cables details.

TABLE 6-15
CABLES PROVIDED BY CATALOG M3090CD

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	MR3/ PD1	MR1	32 (9.76)	966	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminal	Power cable, #8 AWG 5 wire
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wire
90	MR3/ PD1	OW1	51 (15.5)	969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
				048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-29 M3090CH – 3.X-5.4 to LX with ACGD Cable Upgrade for Transportable/ Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-30 M3090CJ – 5.5 Software Site to LX with ACGD Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site

Catalog M3090CK provides fixed site system interconnect cables for Signa Advantage 4.X configuration to Signa Horizon LX with ACGD. Refer to Table 6-16 for cables details.

TABLE 6-16
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	32 (9.75)	968	0.92 (23.4)	2.22x3.84 (56.4x97.6)	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	32 (9.76)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	702	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
				708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	
				774	0.44 (11.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
23	MR2	PP1	41 (12.50)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (51 mm).
(Continued)								

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-16 (Continued)
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				772	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				773	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
45	PP1	MG3	55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm).
				pneumatic tubing	0.25 (6.4)	pneumatic tubing	pneumatic tubing	Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS Pneumatic Patient Alert connection, refer to Group 80 for route.
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
(Continued)								

6-2-31 M3090CK – 4.X To LX With ACGD Cable Upgrade For Fixed Site (Continued)

TABLE 6-16 (Continued)
CABLES PROVIDED BY CATALOG M3090CK

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	MR3/ PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-32 M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site

Catalog M3090CM provides fixed site system interconnect cables for 5.X Signa Horizon configuration to Signa LX with ACGD Gradient. Also included in this catalog are LX upgrade power and ground cables. Refer to Table 6-17 for cables details.

TABLE 6-17
CABLES PROVIDED BY CATALOG M3090CM

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
6	PD1	MR2	32 (9.75)	968	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	32 (9.76)	038	0.188 (4.78)	Hard Wired	Hard Wired	#10 AWG / 1 wire Ground cable
				966	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
				967	0.92 (23.4)	Ring Terminal	Ring Terminal	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
22	MR2	MR1	24 (7.31)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
25	MR1	PP1	21 (6.40)	044	0.20 (5.1)	1.32x0.23 (33.5x5.8)	1.32x0.23 (33.5x5.8)	
				768	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				769	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				770	0.31 (7.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				771	0.415 (10.5)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				775	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				776	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				777	0.212 (5.4)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				783	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
				785	0.64 (16.3)	1.60x2.00 (40.6x50.8)	Ring Terminals	
(Continued)								

6-2-32 M3090CM – 5.X To LX with ACGD Cable Upgrade for Fixed Site (Continued)

TABLE 6-17 (Continued)
CABLES PROVIDED BY CATALOG M3090CM

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG3	55 (16.76) 55 (16.76)	711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend radius of 2 in. (51 mm). Refer to Section 6-8, REAR PEDESTAL AND ENCLOSURE CABLE ACCESS
46	MR1	MR3	59 (17.98)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is a flexible conduit containing fiber optic cables.
80	PA1	MG2	97 (29.6) minus takeup at PA1	—	0.25 (6.4)	pneumatic tubing	pneumatic tubing	This pneumatic tubing is continuously routed from PA1 through PP1 and MG3 to MG3.
87	PP1	MG6	32 (9.76) minus takeup at MG6, See Notes	045	0.195 (4.95)	1/32X0.23 (33.5X5.8)	1/32X0.23 (33.5X5.8)	The usable length is based on MG6 mounted on floor and floor routing of cables.
89	PD1	WC1	67 (20.3)	970	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provides coil cooling water
90	PD1	OW1	51 (15.5)	048	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire, #10 AWG 1 wire
				969	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires
91	MR2	OW1	51 (15.5)	789	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	OW1 end of cable has 3 connectors, each connector plug pull dimensions are 2.3x2.0 (57.2x50.8).
				791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				818	0.44 (11.2)	2.30x2.00 (57.2x50.8)	See Notes	
				836	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
92	PP1	OW1	53 (16.1)	788	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-2-33 M3090CN – 5.5 To LX Cable Upgrade for Transportable/Relocatable (T/R) with Rear Mounted Penetration Panel

All signal cables are replaced during a Transportable or Relocatable Mobile Horizon LX upgrade. PDU to cabinet power cables are replaced but signal cables are re-labelled and re-routed as necessary. Upgrade plans are available from the van manufacturer.

6-2-34 M3090EB – 1.5T EXCITE Hardware Kit

Catalog M3090EB provides additional cables for EXCITE Technology hardware upgrade. Refer to Table 6-18 for cables details.

TABLE 6-18
CABLES PROVIDED BY CATALOG M3090EB

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	1034	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1034 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1025	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1026	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1027	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1028	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				1029	0.34 (8.6)	1.6x2.00 (40.6x50.8)	1.6x2.00 (40.6x50.8)	
				1030	0.525 (13.3)	2.8x2.00 (71.1x50.8)	2.8x2.00 (71.1x50.8)	
				1045	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 1045 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
(Continued)								

6-2-34 M3090EB – 1.5T EXCITE Hardware Kit (Continued)

TABLE 6-18 (Continued)
CABLES PROVIDED BY CATALOG M3090EB

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
45	PP1	MG2/3	30 (9.15)	491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1036	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	
				1037	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1038	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1039	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1040	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				1041	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				1042	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
91	MR2	OW1	51 (15.5)	791	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				1031	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	
				1044	0.34 (8.64)	0.50x0.75 (12.7x19.1)	0.50x0.75 (12.7x19.1)	

6-3 ADVANTAGE WORKSTATION INTERCONNECTS

Refer to Direction 2261309-100 *Advantage Workstation Ultra60 Pre-Installation* for AW interconnect information.

6-4 INTERCONNECTS RE-ROUTED BY HORIZON LX WITH ACGD UPGRADES

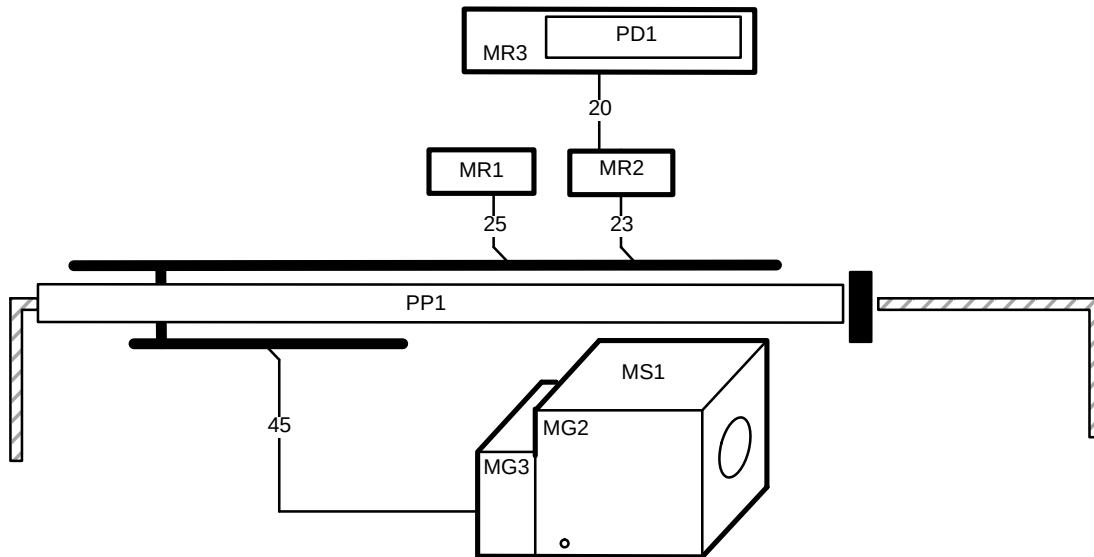
Upgrades to Signa Horizon LX with ACGD replaces the existing system PDU (Full-size, Compact, or PDU in RF/PDU Cabinet) with the PDU located in the lower portion of the ACGD/PDU Cabinet. Therefore, all connections to the PDU may need to be re-routed depending on site layout of the upgraded system.

Note

GE Installation Specialist & site Field Engineer need to consider impact of upgrade site layout on re-routed cables especially if any cables (e.g. power cables) were cut to length for current installed system.

6-5 INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES

Illustration 6-3 and Table 6-19 lists the existing Signa LX system system cables which are removed and discarded during upgrade to Signa LX With EXCITE Technology configuration. If upgrade to ACGD/PDU Cabinet is also being performed then refer to Section 6-6 **INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES** for additional removed and discarded interconnects.



1.5T SYSTEM GROUP INTERCONNECT DIAGRAM
ILLUSTRATION 6-3

6-5 INTERCONNECTS REMOVED BY LX WITH EXCITE TECHNOLOGY UPGRADES (Continued)

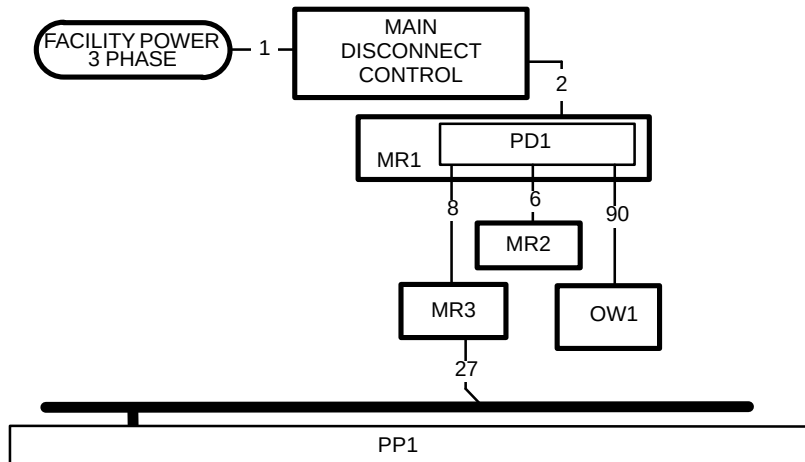
TABLE 6-19
SIGNA LX SYSTEM CABLES REMOVED BY EXCITE TECHNOLOGY UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
20	MR2	MR3	14 (4.27)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (50 mm).
23	MR2	PP1	31 (9.45)	488	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				489	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				490	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				499	0.234 (5.94)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	Used for system with Phased Array only.
				711/ 712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711/712 is a flexible conduit containing fiber optic cables with a minimum bend of 2 in. (50 mm).
25	MR1	PP1	21 (6.40)	486	0.36 (9.1)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Used for system with Phased Array only.
				487	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
				726	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
45	PP1	MG2/3	30 (9.15)	260/ 700	0.23 (5.94)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Used for system with Phased Array only.
				491	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				492	0.36 (9.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				493	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				494	0.24 (6.1)	0.55x1.10 (14.0x27.9)	0.55x1.10 (14.0x27.9)	Used for system with Phased Array only.
				495	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.
				727	0.45 (11.4)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	Used for system with Phased Array only.

6-6 INTERCONNECTS REMOVED BY HORIZON LX WITH ACGD UPGRADES

6-6-1 Signa Horizon LX System With SGD & RF/PDU Cabinet

Illustration 6-4 and Table 6-20 lists the existing Signa Horizon LX system with SGD & RF/PDU Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration.



SIGNA HORIZON LX WITH SGD & RF/PDU CABINET TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-4

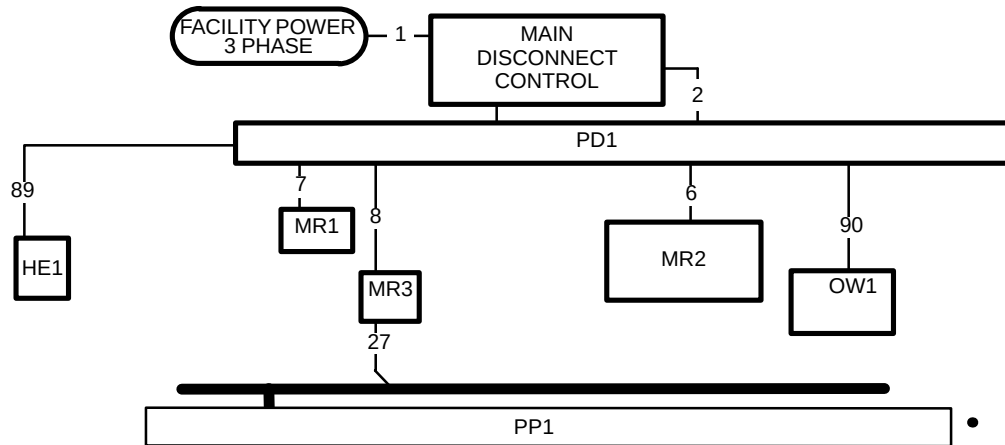
TABLE 6-20

SIGNA HORIZON LX WITH SGD & RF/PDU CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-2 Signa Horizon LX System With SGD & RF/Penetration Cabinet

Illustration 6-5 and Table 6-21 lists the existing Signa Horizon LX system with SGD & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration.



SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-5

TABLE 6-21
SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.

6-6-2 Signa Horizon LX System With SGD & RF/Penetration Cabinet (Continued)

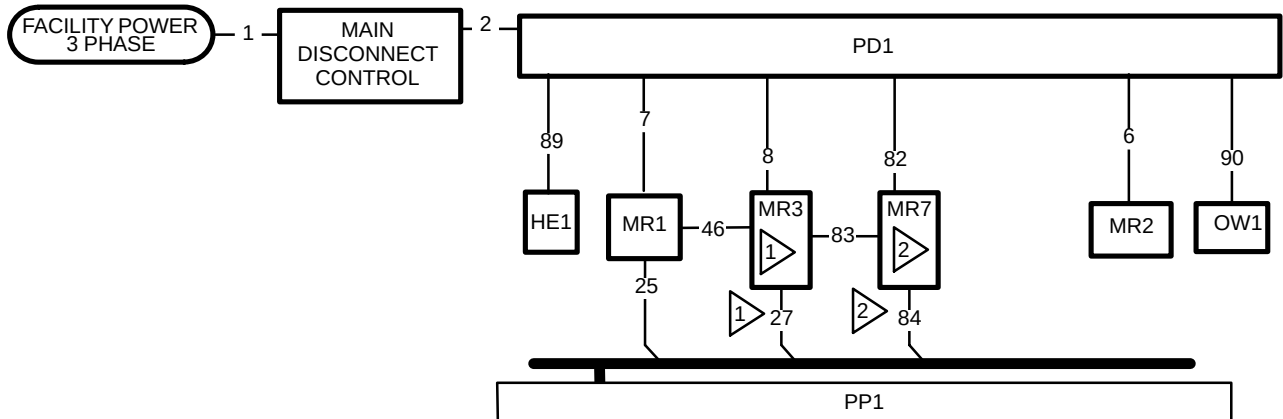
TABLE 6-21 (Continued)

SIGNA HORIZON LX WITH SGD & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
27	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

6-6-3 Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet

Illustration 6-6 and Table 6-22 lists the existing Signa Horizon LX system with 8645 Gradients & RF/Penetration Cabinet system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration. Base system configuration used 8645 Gradient Cabinet. HiSpeed and EchoSpeed system configurations used 8645 Gradient Cabinet and GRAM Cabinet.



1 BASE SYSTEM CONFIGURATION USED 8645 GRADIENT CABINET (MR3) & CABLE GROUP 27 RUNS.

2 HISPEED & ECHOSPEED SYSTEM CONFIGURAITONS USED 8645 GRADIENT CABINET (MR3), GRAM CABINET (MR7), AND CABLE GROUPS 82, 83, & 84 RUNS. THESE RUNS ARE REMOVED ONLY IF GRAM CABINET (MR7) IS PART OF THE EXISTING SYSTEM BEING UPGRADED.

**SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENENTRATION CABINET TO ACGD UPGRADES
REMOVED INTERCONNECTS DIAGRAM**

ILLUSTRATION 6-6

TABLE 6-22

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
6	MR3/ PD1	MR2	22 (6.70)	030	0.92 (23.4)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.

6-6-3 Signa Horizon LX System With 8645 Gradients & RF/Penetration Cabinet (Continued)

TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

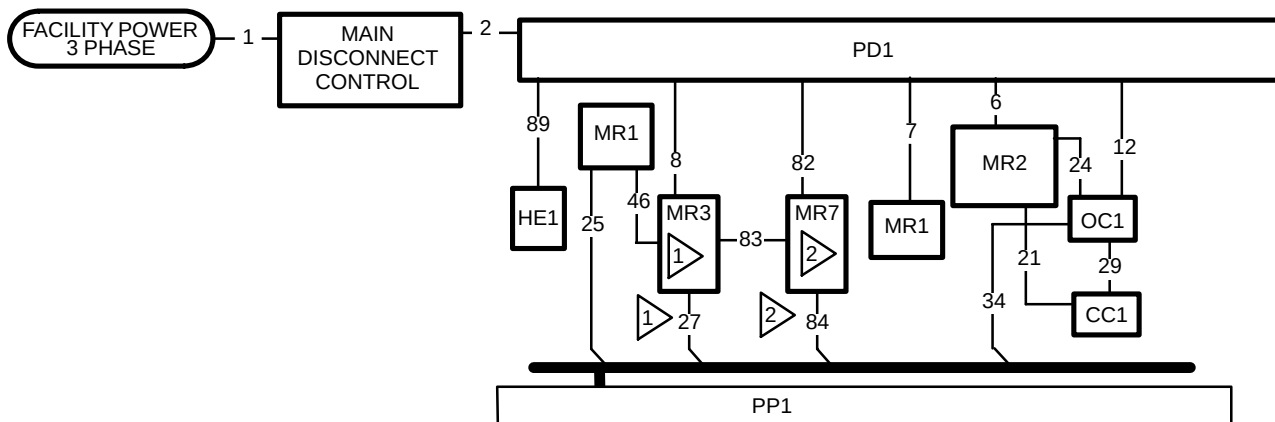
GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
27 See Note 1	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
82 See Note 2	PD1	MR7	22 (6.70)	043	0.92 (23.4)	Ring Terminals	2.50x4.00 (63.5x102)	Power cable, #8 AWG 5 wires
83 See Note 2	MR1	MR7	9 (4.6)	753	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				754	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				755	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				756	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				757	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				758	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				759	1.64 (41.7)	Ring Terminals	Ring Terminals	
				760	1.64 (41.7)	Ring Terminals	Ring Terminals	
				761	1.64 (41.7)	Ring Terminals	Ring Terminals	
84 See Note 2	MR7	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
90	PD1	OW1	51 (15.5)	047	0.70 (17.78)	Ring Terminals		Power Cable, #10 AWG 4 wires

Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs.

2 HiSpeed & EchoSpeed system configuraitons used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed onlu if GRAM Cabinet (MR7) is part of the existing system being upgraded.

6-6-4 Signa Horizon 5.X System

Illustration 6-7 and Table 6-23 lists the existing Signa Horizon 5.X system system cables which are removed and discarded during upgrade to Signa Horizon LX with ACGD configuration. Base system configuration used 8645 Gradient Cabinet. HiSpeed and EchoSpeed system configurations used 8645 Gradient Cabinet and GRAM Cabinet.



- 1 BASE SYSTEM CONFIGURATION USED 8645 GRADIENT CABINET (MR3) & CABLE GROUP 27 RUNS.
- 2 HISPEED & ECHOSPEED SYSTEM CONFIGURAITONS USED 8645 GRADIENT CABINET (MR3), GRAM CABINET (MR7), AND CABLE GROUPS 82, 83, & 84 RUNS. THESE RUNS ARE REMOVED ONLY IF GRAM CABINET (MR7) IS PART OF THE EXISTING SYSTEM BEING UPGRADED.

SIGNA HORIZON 5.X TO ACGD UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-7

TABLE 6-23
SIGNA HORIZON 5.X SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be replaced to meet Section 5, Power Requirements.
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	22 (6.70)	041	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #4 AWG 5 wires
				042	1.35 (34.3)	Ring Terminals	Ring Terminals	Run 42 is used only for 2 bay 8645 Gradient Cabinet configuration.
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires

6-6-4 Signa Horizon 5.X System (Continued)

TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
27 See Note 1	MR3	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
82 See Note 2	PD1	MR7	22 (6.70)	043	0.92 (23.4)	Ring Terminals	2.50x4.00 (63.5x102)	Power cable, #8 AWG 5 wires
Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs. 2 HiSpeed & EchoSpeed system configuraitons used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed onlu if GRAM Cabinet (MR7) is part of the existing system being upgraded.								

6-6-4 Signa Horizon 5.X System (Continued)

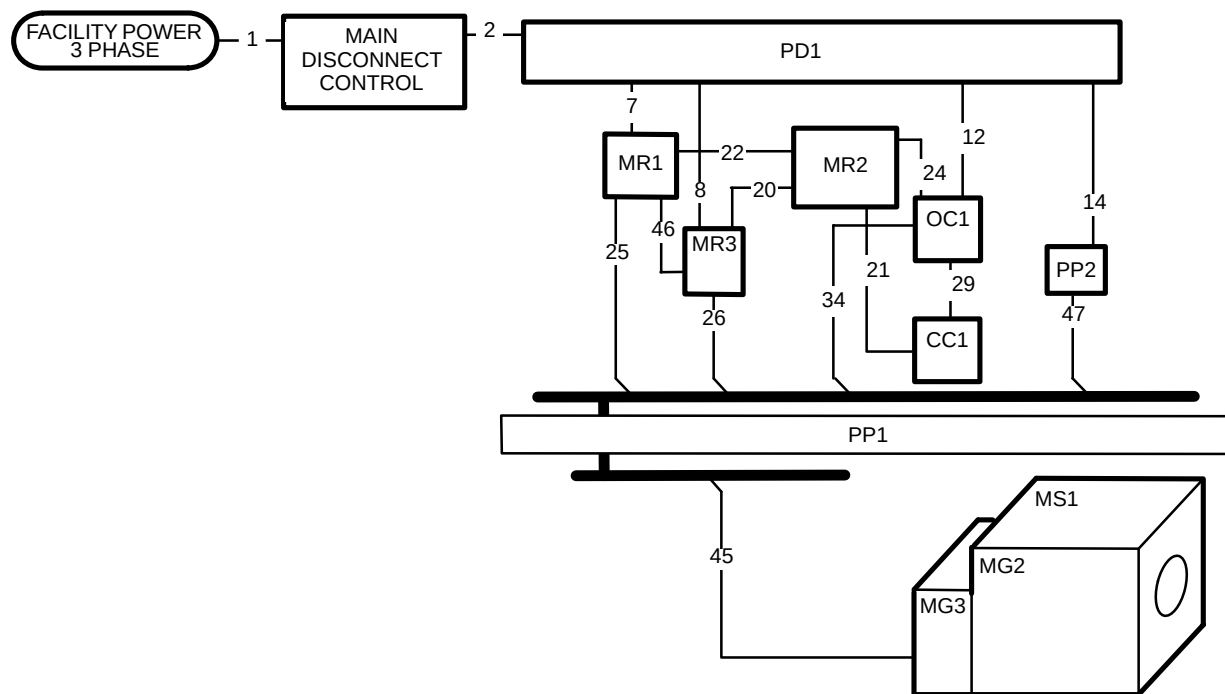
TABLE 6-22 (Continued)

SIGNA HORIZON LX WITH 8645 GRADIENTS & RF/PENETRATION CABINET SYSTEM CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
83 See Note 2	MR1	MR7	9 (4.6)	753	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				754	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				755	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				756	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				757	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				758	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				759	1.64 (41.7)	Ring Terminals	Ring Terminals	
				760	1.64 (41.7)	Ring Terminals	Ring Terminals	
				761	1.64 (41.7)	Ring Terminals	Ring Terminals	
84 See Note 2	MR7	PP1	41 (12.5)	762	1.21 (30.7)	Ring Terminals	Ring Terminals	Magnet Room side of these runs are not replaced during the upgrade.
				763	1.21 (30.7)	Ring Terminals	Ring Terminals	
				764	1.21 (30.7)	Ring Terminals	Ring Terminals	
89	PD1	HE1	67 (20.3)	046	0.635 (16.1)	Hard Wired	Hard Wired	Mobile configurations do not use WC1, van chiller system provided coil cooling water
Note 1 Base system configuration used 8645 Gradient Cabinet (MR3) & cable group 27 Runs. 2 HiSpeed & EchoSpeed system configuraitons used 8645 Gradient Cabinet (MR3), GRAM Cabinet (MR7), and cable groups 82, 83, & 84 Runs. These Runs are removed onlu if GRAM Cabinet (MR7) is part of the existing system being upgraded.								

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System

Illustration 6-8 and Table 6-24 lists the existing 5.1 to 5.4 Signa Advantage and 5.5 software only system cables which are removed and discarded during upgrade to Horizon LX with ACGD configurations.



5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE UPGRADES REMOVED INTERCONNECTS DIAGRAM
ILLUSTRATION 6-8

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

TABLE 6-24

5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY ACGD UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	31 (9.45)	003	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
20	MR2	MR3	24 (7.31)	710	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 710 is flexible conduit containing fiber optic cable(s) with a minimum bend of 2 in. (51 mm).
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
22	MR2	MR1	24 (7.32)	708	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 708 is flexible conduit containing fiber optic cables.
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
25	MR1	PP1	41 (12.5)	470	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
(Continued)								

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

TABLE 6-24 (Continued)

5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADES

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
26	MR3	PP1	33 (10.0)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	These runs were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 45.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
45	PP1	MG3	40 (12.2)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 26.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
				412	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				474	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				475	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				476	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				477	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				478	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
				711	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 711 is flexible conduit containing fiber optic cables.
				712	1.04 (26.4)	1.04x2.00 (26.4x50.8)	1.04x2.00 (26.4x50.8)	Run 712 is flexible conduit containing fiber optic cables.
				(Continued)				

6-6-5 Signa Horizon 5.X Software Only & Signa Advantage 5.X System (Continued)

TABLE 6-24 (Continued)

5.5 SOFTWARE ONLY SITE & 5.1-5.4 SIGNA ADVANTAGE CABLES REMOVED BY HORIZON LX UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
46	MR1	MR3	24 (7.32)	709	0.83 (21.1)	0.83x2.00 (21.1x50.8)	0.83x2.00 (21.1x50.8)	Run 709 is flexible conduit containing fiber optic cables.
47	PP1	PP2	21 (6.4)	384	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				386	0.53 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				388	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				389	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				413	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	

6-6-6 Signa Advantage 4.X System

Illustration 6-9 and Table 6-25 lists the existing Signa Advantage 4.X system cables which are removed and discarded during upgrade to Horizon LX with ACGD configurations. Note, all Signa Advantage 4.X system configurations are covered, including both forward product and runs provided by various 4.X system upgrades.

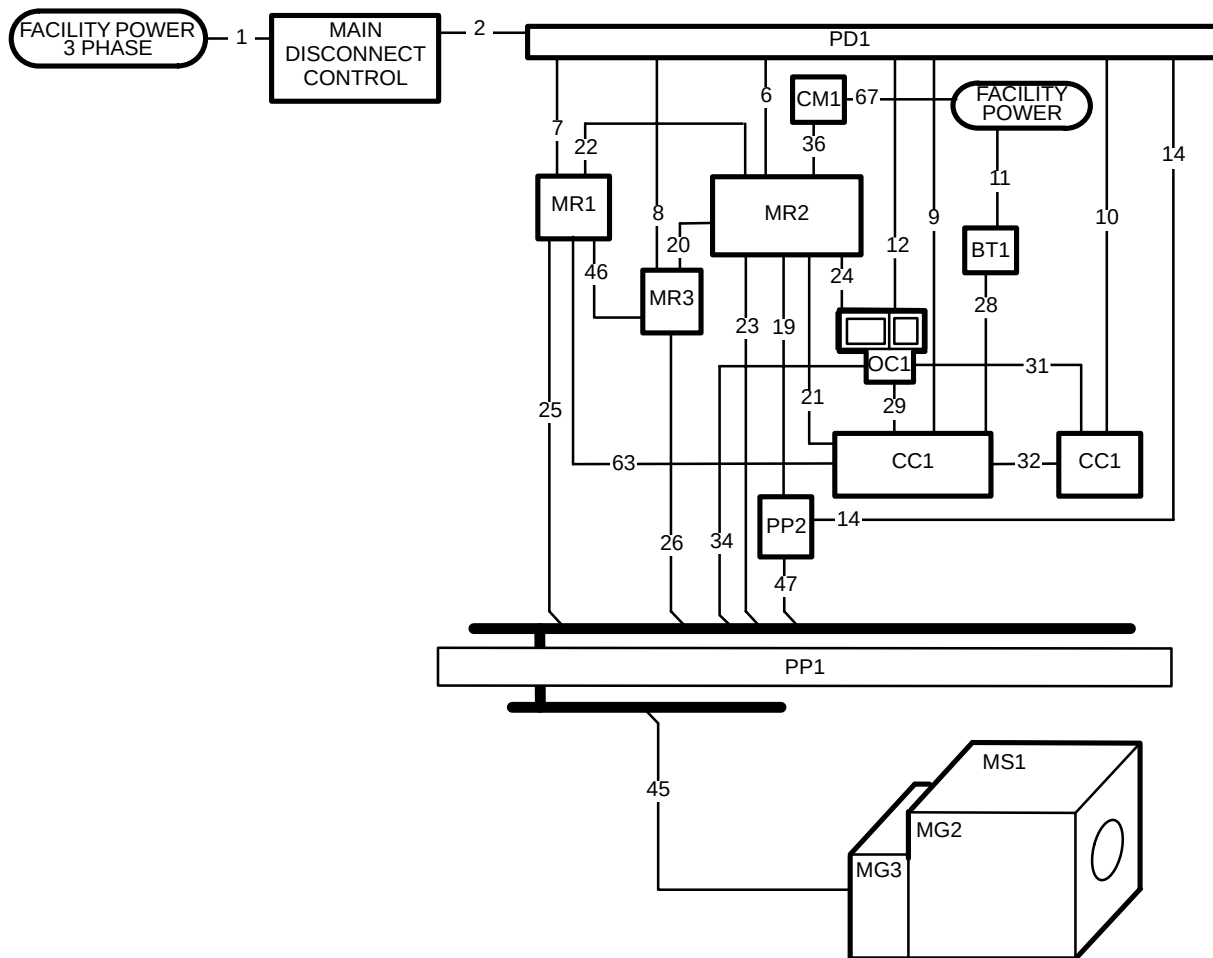
**SIGNA ADVANTAGE 4.X UPGRADES REMOVED INTERCONNECTS DIAGRAM**

ILLUSTRATION 6-9

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
1	Facility Power	Main Disc.	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
2	Main Disc.	PD1	—	—	—	—	—	Customer Furnished, may need to be placed to meet Section 5, Power Requirements.
6	PD1	MR2	32 (9.75)	002	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
7	PD1	MR1	22 (6.70)	009	0.92 (23.4)	Ring Terminal	Ring Terminal	Power Cable, #8 AWG 5 wires
8	PD1	MR3	31 (9.45)	003	1.35 (34.3)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
9	PD1	CC1	32 (9.75)	014	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				015	0.92 (23.4)	Ring Terminals	2.25x3.90 (57.2x99.1)	Power cable, #8 AWG 5 wires
				021	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
10	PD1	CC2	32 (9.75)	016‡ or 035**†	0.92 (23.4)	Ring Terminals	2.25x3.90 (57.2x99.1)	Power cable, #8 AWG 5 wires
				017**†	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				022**†	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
11	Facility Power	BT1	—	—	—	—	—	Customer Furnished.
12	PD1	OC1	52 (15.8)	012	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				013	0.635 (16.1)	Ring Terminals	2.00x4.30 (50.8x109)	Power cable, #12 AWG 3 wires
				023	0.188 (4.78)	Ring Terminals	Ring Terminals	Ground wire #10 AWG 1 wire
14	PD1	PP2	32 (9.75)	027	0.635 (16.1)	Ring Terminals	2.40x5.00 (61x127)	Power cable, #12 AWG 3 wires
				028	0.850 (21.6)	Ring Terminals	2.03x4.30 (51.6x109)	Power cable, #8 AWG 3 wires
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
19	MR2	PP2	44 (13.4)	417	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
20	MR2	MR3	24 (7.32)	200	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				201	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				202	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				203	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
21	MR2	CC1	59 (17.98)	228 †	0.42 (10.7)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				427	0.42 (10.7)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
				428 or 448**	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				430 or 449**	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				431 or 450**	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
22	MR2	MR1	24 (7.32)	226	0.35 (8.90)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
23	MR2	PP1	41 (12.5)	419	0.30 (7.62)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	Not present with Run 454.
				425	0.525 (13.3)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Not present with Run 425.
				454	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
24	MR2	OC1	64 (19.5)	287	0.35 (8.9)	1.30x2.00 (33.5x50.8)	2.80x2.00 (70.4x50.8)	
25	MR1	PP1	41 (12.5)	416 ‡	0.212 (5.38)	0.57x2.00 (14.5x50.8)	0.57x2.00 (14.5x50.8)	
				470**	0.34 (8.64)	1.60x2.00 (40.6x50.8)	1.60x2.00 (40.6x50.8)	
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
26	MR3	PP1	33 (10.0)	405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 45.
				406	0.70 (17.8)	Ring Terminals	Ring Terminals	
				407	0.70 (17.8)	Ring Terminals	Ring Terminals	
28	CC1	BT1	44 (13.4)	244 † or 429	0.464 (11.8)	2.10x2.10 (53.3x53.3)	2.10x2.10 (53.3x53.3)	
29	CC1	OC1	74 (22.6)	238	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				239	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				433	0.41 (10.4)	0.60x1.90 (15.2x48.3)	0.60x1.90 (15.2x48.3)	
				434	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
31	CC2	OC1	74 (22.6)	248 †	0.41 (10.4)	0.60x1.90 (15.2x48.3)	0.60x1.90 (15.2x48.3)	
				249 †	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
32	CC2	CC1	9 (2.74)	287**	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				301**	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				302**	0.5 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				325**	1.0 (25.4)	2.70x2.40 (68.6x60.1)	2.70x2.40 (68.6x60.1)	
				327**	1.0 (25.4)	2.70x2.40 (68.6x60.1)	2.70x2.40 (68.6x60.1)	
34	OC1	PP1	56 (17.1)	256	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46-306900) only. ‡ These runs are applicable to 4.0 to pre-4.5 Signa Advantage systems (have RF Cabinet Model 46-282900) only.								
(Continued)								

6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES	
36	MR2	CM1	73 (22.3)	346**	0.25 (6.35)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
				347**	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
45	PP1	MG3	53 (16.2)	261	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Not present with Run 453.	
				405	0.70 (17.8)	Ring Terminals	Ring Terminals	Runs 405, 406, and 407 were cut and connected to PP1 at site: total length of 100 ft (30.5 m) included both Equipment and Magnet Rooms. Refer to Group 26.	
				406	0.70 (17.8)	Ring Terminals	Ring Terminals		
				407	0.70 (17.8)	Ring Terminals	Ring Terminals		
				411	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
				412	0.35 (8.9)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)		
				415 ‡ or 476**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
				420	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)		
				424	0.18 (4.57)	0.3x1.0–X2 (7.62x25.4)	0.3x1.0–X2 (7.62x25.4)		
				446 ‡	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
				447 ‡	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
				453	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)		Not present with Run 261.
				474**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
				475**	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)		
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46–306900) only. ‡ These runs are applicable to 4.0 to pre–4.5 Signa Advantage systems (have RF Cabinet Model 46–282900) only.									
(Continued)									

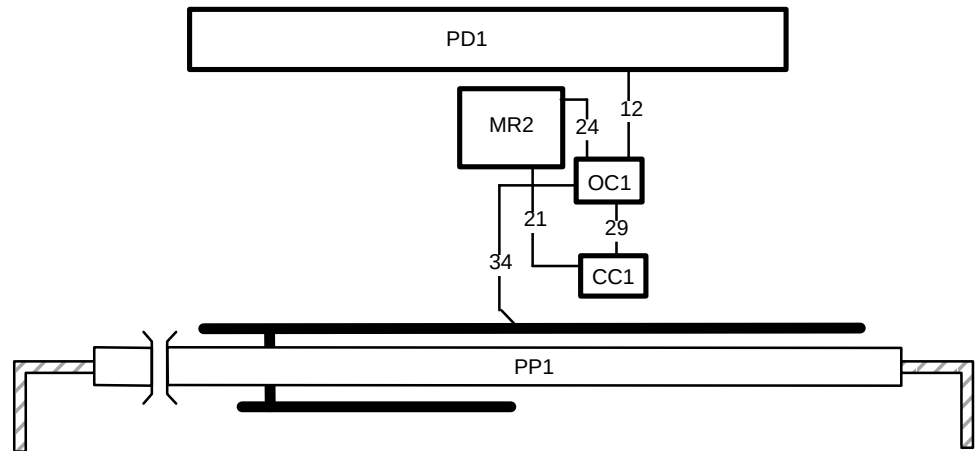
6-6-6 Signa Advantage 4.X System (Continued)

TABLE 6-25 (Continued)
4.X SIGNA ADVANTAGE CABLES REMOVED BY HORIZON UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
46	MR1	MR3	59 (17.98)	426	0.525 (13.3)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
47	PP1	PP2	21 (6.4)	384	0.525 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				386	0.53 (13.3)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				388	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
				389	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				410	0.525 (12.7)	2.80x2.00 (70.4x50.8)	2.80x2.00 (70.4x50.8)	
				413	0.36 (9.14)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	
63	MR1	CC1	59 (17.98)	224	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				274	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
67	Facility Power	CM1	—	—	—	—	—	Customer Furnished.
Note ** These runs may be present in systems that were upgraded to 4.X Signa Advantage systems only, not present in forward production 4.X Signa Advantage systems. † These runs are applicable to 4.5 to 4.8 Signa Advantage systems (have RF Cabinet Model 46-306900) only. ‡ These runs are applicable to 4.0 to pre-4.5 Signa Advantage systems (have RF Cabinet Model 46-282900) only.								

6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE

Illustration 6-10 and Table 6-26 lists the existing Signa Horizon 5.X system cables which are removed and discarded during upgrade to Signa Horizon LX Computer only configuration with no gradient subsystem change.



SIGNA HORIZON 5.X UPGRADE REMOVED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-10

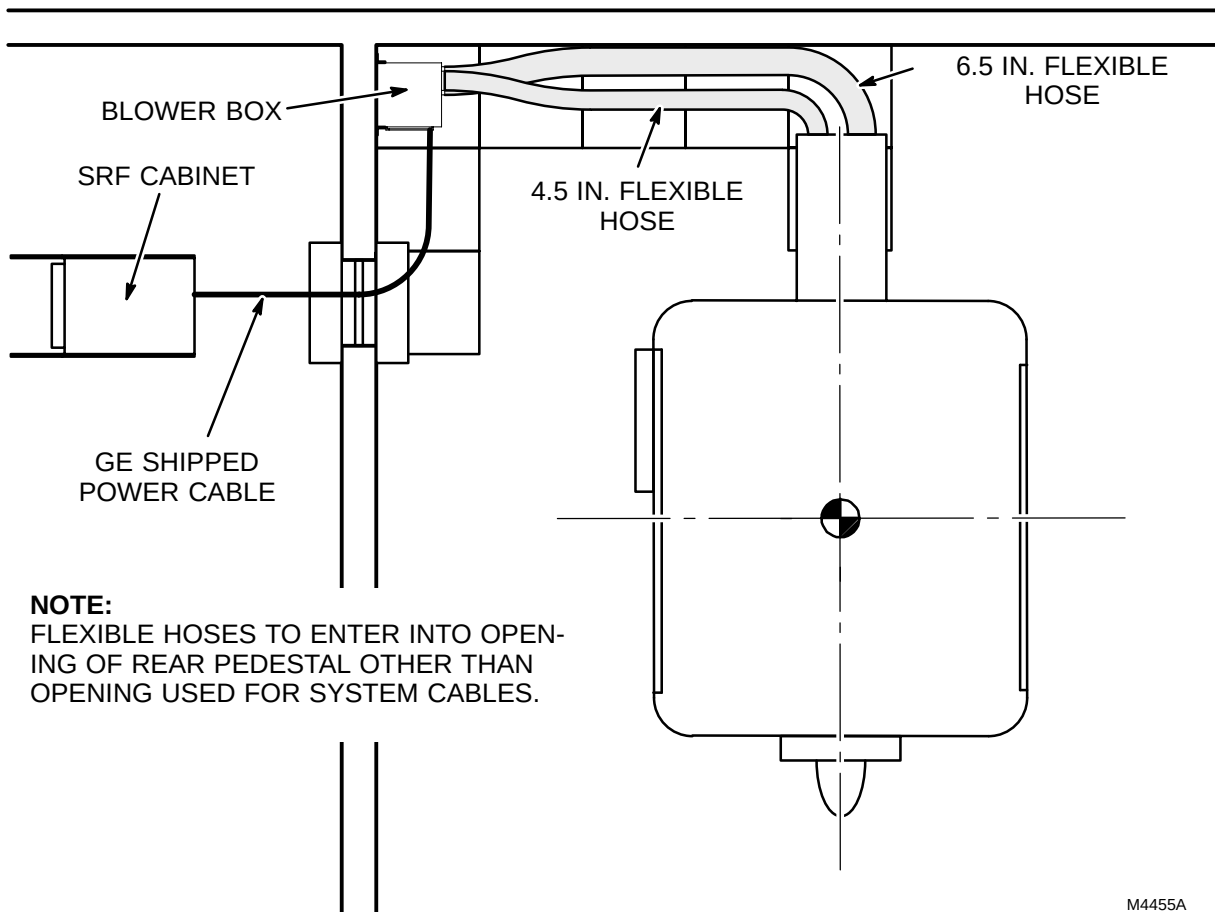
**6-7 INTERCONNECTS REMOVED BY SIGNA HORIZON 5.X TO SIGNA LX COMPUTER ONLY UPGRADE
(Continued)**

TABLE 6-26
SIGNA HORIZON 5.X CABLES REMOVED BY HORIZON LX COMPUTER ONLY UPGRADE

GROUP NUMBER	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
	FROM	TO				FROM	TO	
12	PD1	OC1	52 (15.8)	023	0.188 (4.78)	Ring Terminal	Ring Terminal	Ground wire, #10 AWG 1 wire
				050	1.02 (25.9)	Ring Terminals	Ring Terminals	Power cable, #8 AWG 5 wires
21	MR2	CC1	11 (3.35)	064	0.78 (19.8)	2.00x3.75 (50.8x95.3)	2.00x3.75 (50.8x95.3)	Run 064 is power for CC1 from MR2.
				719	0.625 (15.9)	3.00x2.00 (76.2x50.8)	3.00x2.00 (76.2x50.8)	
				787	0.48 (12.2)	2.30x2.00 (58.4x50.8)	2.30x2.00 (58.4x50.8)	
24	MR2	OC1	64 (19.51)	707	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				718	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
29	CC1	OC1	74 (22.56)	800	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				801	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
				802	0.43 (10.9)	1.60x1.90 (40.6x48.3)	1.60x1.90 (40.6x48.3)	
				803	0.242 (6.15)	0.57x1.07 (14.5x27.2)	0.57x1.07 (14.5x27.2)	
				804	0.48 (12.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
34	OC1	PP1	56 (17.1)	704	0.464 (11.8)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	

6-8 REAR PEDESTAL AND ENCLOSURE CABLE ACCESS

Signa Horizon has an increase of cabling and hoses routed through the Rear Pedestal cable access areas. These include the Blower Box, 2 flexible hoses, 3 new large Gradient cables, 2 water lines for Body Coil, and the remainder of the Magnet Enclosure interconnect cables. Rear Pedestal cable access and routing needs to be considered. Sites with cable access from under the Rear Pedestal should have enough cable access area. Sites using Rear Pedestal cable access on rear or one side will need to plan to utilize more than one access opening. Illustration 6-11 shows one example of Blower Box set-up with flexible hoses route separate from other interconnects.

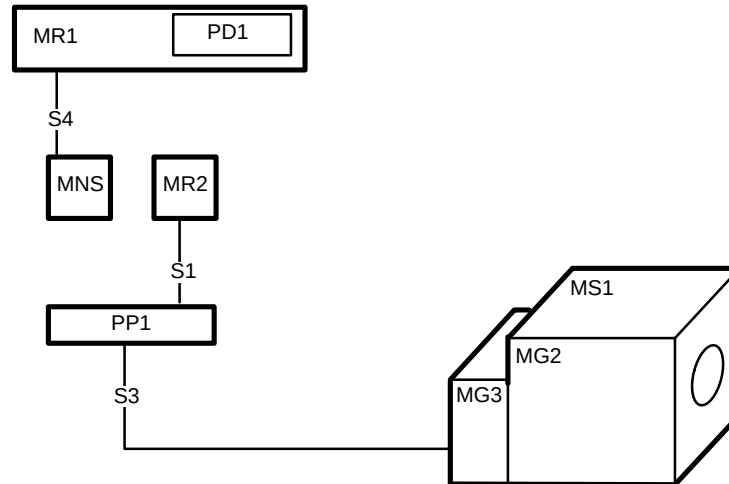


EXAMPLE OF SIGNA HORIZON BLOWER BOX SET-UP
ILLUSTRATION 6-11

6-9 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH ACGD

6-9-1 M3090DA – 4.X-5.4 MNS Upgrade Kit For ACGD Site

Illustration 6-12 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-27 contains the cable data for these interconnects.



CATALOG M3090DA, M3090DB, & M3090DC SPECTROSCOPY OPTION SYSTEM INTERCONNECTS
ILLUSTRATION 6-12

TABLE 6-27
CABLES PROVIDED BY CATALOG M1033MD, M3090DA, M3090DB, & M3090DC

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	

6-9-2 M3090DB – 5.5 MNS Upgrade Kit For ACGD Site

Cables provided by catalog M3090DB are the same as catalog M3090DA. Refer to Illustration 6-12 and Table 6-27 cables details.

6-9-3 M3090DB – LX MNS Upgrade Kit For ACGD Site

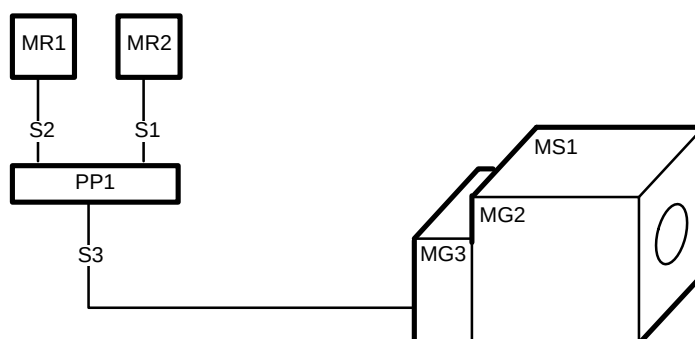
Cables provided by catalog M3090DC are the same as catalog M3090DA. Refer to Illustration 6-12 and Table 6-27 cables details.

6-10 SPECTROSCOPY OPTION INTERCONNECTS FOR 1.5T SYSTEM WITH SGD

Catalogs M1033MA and M1033MB utilize the already installed Spectroscopy option system interconnect cables. Catalogs M1033MC and M1033MD supply additional system interconnect cables, refer to the following sub-sections for details.

6-10-1 M1033MC – MNS Kit For Signa LX (8.X) Systems That Do Not Have RF/PDU Cabinet

Illustration 6-13 shows the 1.5T system group interconnects for the Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by M1033MC. Table 6-28 contains the cable data for these interconnects.



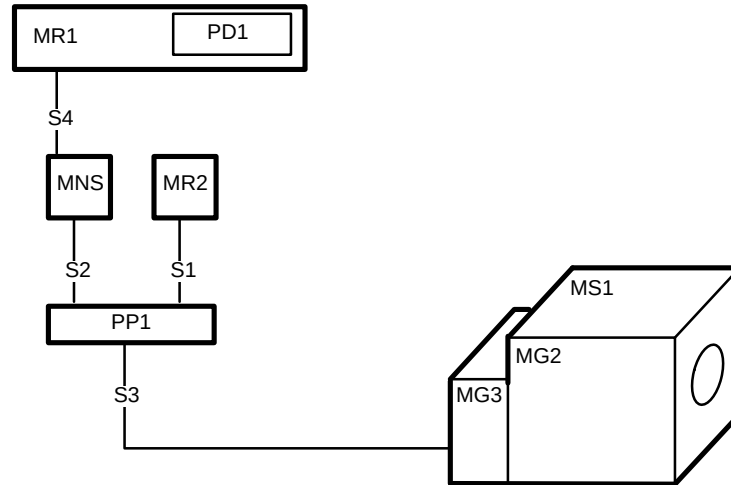
**CATALOG M1033MC SPECTROSCOPY OPTION SYSTEM INTERCONNECTS
ILLUSTRATION 6-13**

**TABLE 6-28
CABLES PROVIDED BY CATALOGS M1033MC**

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MR1	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	

6-10-2 M1033MD – MNS Kit For Signa LX (8.X) Systems That Have RF/PDU Cabinet

Illustration 6-14 shows the 1.5T system group interconnects for the MNS Cabinet and Spectroscopy Modules added to the Penetration Panel and Magnet Enclosure by the spectroscopy upgrade options listed below. Table 6-29 contains the cable data for these interconnects.



CATALOG M1033MD SPECTROSCOPY OPTION SYSTEM INTERCONNECTS

ILLUSTRATION 6-14

TABLE 6-29
CABLES PROVIDED BY CATALOG M1033MD

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
S1	0.035 (22.8)	MR2	PP1	31 (9.45)	469	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnect to Table 6-3 Group 23.
S2	0.41 (267)	PP1	MNS	21 (6.40)	468	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	Additional interconnects to Table 6-3 Group 25.
S3	0.357 (231)	PP1	MG3	30 (9.15)	472	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	Additional interconnects to Table 6-3 Group 45.
					473	0.64 (16.3)	0.80x2.50 (20.3x63.5)	0.80x2.50 (20.3x63.5)	
S4	0.656 (425)	MR1	MNS	9 (2.74)	936	0.68 (17.3)	2.00x4.00 (50.8x102)	2.00x3.63 (50.8x92.2)	Cabinet Power Cable, #10 AWG 3 wires
					961	0.44 (11.2)	2.30x2.00 (57.2x50.8)	2.30x2.00 (57.2x50.8)	
					962	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					963	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					964	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	
					965	0.212 (5.38)	0.57x1.125 (14.5x28.6)	0.57x1.125 (14.5x28.6)	